Wage Scars and Life-Cycle Dynamics: A Portable Income Process*

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Abstract

Involuntary job separation causes earnings to fall 15.4% initially and remain depressed two decades later. We quantify how much these scars shape the income distribution: incorporating the empirical separation hazard and scar profile into a standard income process, we find that job loss reduces average life-cycle income growth by 14.7% and raises cross-sectional dispersion by 17.8% relative to a counterfactual without separations. We then highlight a simple but underappreciated implication of life-cycle concavity for reduced-form income processes: any specification in which displaced workers restart along the same concave path mechanically delivers rapid recovery, missing the persistent scar in the data. We use a parsimonious learning-by-doing framework to show that this tension cannot be resolved by selection on wages or serially correlated separations alone; some channel that yields lower post-separation income growth is required. We provide a calibrated process matching both scars and life-cycle moments, intended for researchers who need a portable income process with realistic job-loss risk.

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1 Introduction

Involuntary job loss has long-lasting consequences for wages. In the US, hourly earnings initially fall by 15.4% on average following a separation and remain substantially depressed even twenty years later relative to similar workers who do not experience a separation.[1] This finding is robust across countries and time periods. Researchers studying the implications of income risk and wage inequality for the design of social insurance, consumer credit, and many other questions need wage processes that incorporate these scars in a way that is also consistent with what we know about life-cycle wage dynamics. The goal of this paper is to provide one.

We make three contributions. First, we estimate how much job loss shapes the aggregate wage distribution. Embedding empirically estimated separation hazards and scar profiles into a standard wage process, we find that turning off separations would raise average life-cycle wage growth by 14.7% and reduce cross-sectional wage dispersion by 17.8%. Job loss is therefore quantitatively central to life-cycle wage dynamics, not just a tail risk relevant to a small subset of workers.

Second, we draw out a simple implication of life-cycle concavity that constrains what a reduced-form wage process can do. The empirical life-cycle profile of wages is strongly concave: most wage growth occurs in the first several years of experience. If a displaced worker restarts along that same path, then she stands on its steepest portion and mechanically recovers most of her loss within a few years. A persistent gap relative to a never-separated reference group requires that the post-separation path differ from the pre-separation path and specifically, that post-separation growth be lower on net. Figure 1 illustrates this point: a worker who restarts on the baseline path quickly catches up (left); the same worker on a path with 75% of baseline growth carries a persistent scar (right).

[1] We use hourly earnings and wages interchangeably. The literature on "scarring" typically uses "displaced" to refer to high-tenure male workers; we use "separated" to refer to all involuntary

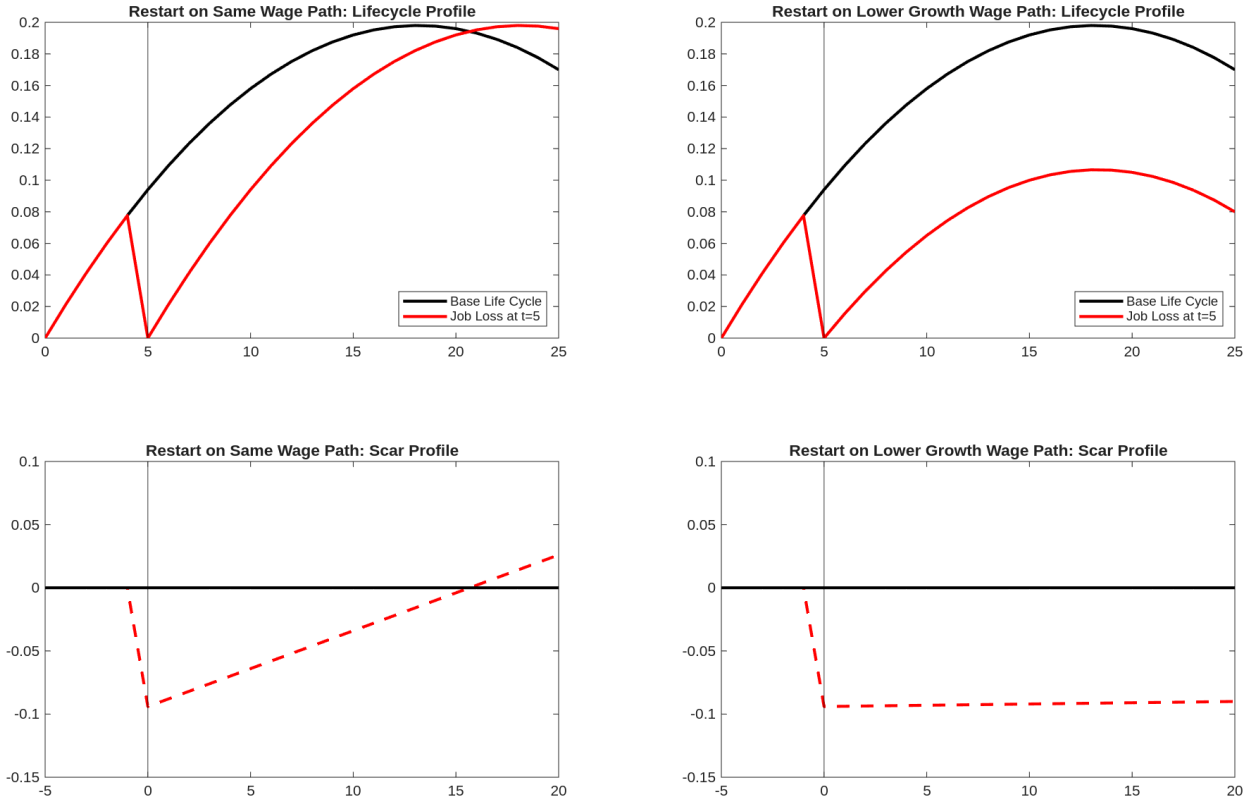

Figure 1: **Restart Path and Scar Persistence.** Top panels show the life-cycle wage profile; bottom panels show the gap relative to a never-separated worker. Left: separated workers restart along the same wage process. Right: separated workers restart along a wage process with 75% of baseline growth.

Third, we provide a calibrated wage process that delivers this. We work within a parsimonious framework of wage dynamics akin to learning-by-doing, and consider four variants: a baseline with random separation and human-capital loss; selection of low-wage workers into separation; serially correlated separations; and lower post-separation wage growth. When models are required to match both empirical scars *and* the life-cycle moments of the never-separated reference group, only the lowered-growth specification succeeds. The first three either understate persistence or distort the life-cycle profile. We report parameter estimates and provide the implied process for researchers to embed in their own work.

separations regardless of pre-separation tenure or experience.

Relation to the structural literature on scarring. A growing structural literature has made important progress in understanding the mechanisms behind wage scars, including Krolikowski (2017), Michaud (2018), Huckfeldt (2022), Jung and Kuhn (2019), Jarosch (2023), and Burdett et al. (2020). The result we draw out is consistent with these papers: each generates persistent scars through channels that, on net, deliver lower post-separation wage growth. In Burdett et al. (2020), forgone learning-by-doing during unemployment accounts for the bulk of lifetime earnings losses. In Huckfeldt (2022), displaced workers sort into job types with lower skill-growth rates. In Michaud (2018), a separation reveals workers to be of low productivity and limits their future opportunities for wage growth. In the others, repeated separation interrupts human-capital accumulation across jobs. These richer mechanisms differ in their economic content, but each produces an outcome that, viewed through the lens of a reduced-form wage process, looks like lower post-separation growth. Our parsimonious analysis is best understood as making this common feature transparent: any successful theory must produce something that, on net, lowers post-separation growth, regardless of whether the underlying mechanism operates through human capital, job ladders, employer match quality, or other channels. Our deliverable, a calibrated wage process that builds in this feature directly, is intended for applied researchers who need realistic income risk including job loss but do not need to take a stand on which structural mechanism produces it.[2]

[2]Our exercise complements Davis and von Wachter (2011), who emphasize that frictional wage dispersion in standard search-matching models, which Hornstein et al. (2011) estimate at around 5% above the lowest wage, is too small to deliver the 15% scars in the data. We instead start from a wage process that does deliver realistic life-cycle dispersion and ask what is required of the displacement process given that.

2 Empirical Wage Scars

We estimate wage scars using the Panel Study of Income Dynamics (PSID) 1976–2015 waves. We follow the standard event-study framework of Jacobson et al. (1993), with sample construction details and identification of involuntary separations following Stevens (1997).[3] We estimate the effect of involuntary separation at time $t-n$ on log real hourly earnings:

$$ln(w_{i,t}) = \Phi \mathbf{X}_{i,t} + \Theta \mathbf{E}_{i,t} + \sum_{n=-2}^{19} \beta_{t-n} D1_{i,t-n} + \gamma_1 D1_{i,20+} + \sum_{j=2}^{5} \gamma_j Dj_i + \delta_t y_t + \zeta_i + \eta_s S + \epsilon_{i,t} \tag{1}$$

The dummy variables $D1_{i,t-n}$ indicate years from the first involuntary separation, with $n \in \{-2, -1, 0, 1, \ldots, 19\}$ and a separate dummy for 20+ years post-separation. Following Stevens (1997), multiple separations are important, so we additionally include dummies Dj_i for whether the worker has had at least $j \in \{2, 3, 4, 5\}$ separations. Controls $\mathbf{X}_{i,t}$ include experience, experience squared, and union participation; $\mathbf{E}_{i,t}$ are education dummies. Year, state, and individual fixed effects are denoted y_t, S, and ζ_i.

Figure 2 (left panel) shows the estimated scar profile from the first separation, $\exp(\beta_{t-n}) - 1$, with 95% confidence intervals.[4] Hourly earnings fall by 15.4% on impact and remain depressed by 7–10% even fifteen years out. These magnitudes are consistent with the broader scarring literature: Davis and von Wachter (2011) report initial losses just over 10% in administrative data, with losses of 5% twenty years

[3] Our sample includes male household heads with at least three years of pre-separation labor force attachment. Involuntary separations are identified from the PSID's reason-for-separation question and exclude quits, retirements, and within-firm transitions. The Appendix provides full details on sample construction, separation identification, and the experience-cleaning procedure of Kambourov and Manovskii (2009). We use hourly wages rather than total earnings to focus on the permanent rather than transitory consequences of job loss; effects on hours are largely transitory in our sample, recovering within three years.

[4] Full coefficient estimates are reported in the Appendix. Losses are computed as $e^{\beta_{t-n}} - 1$.

later.

3 How Much Does Job Loss Shape the Wage Distribution?

We next quantify how much these scars contribute to standard summaries of the wage distribution. We do so by embedding the empirically estimated separation hazards and scar profile into a standard Guvenen (2009)-style wage process and comparing simulated moments under separation to a counterfactual with separations turned off.[5]

We specify the wage process for non-separated workers in standard form:

$$\ln(w_{it}) = \alpha_i + \beta_1 exp + \beta_2 exp^2 + z_{it} + \epsilon_{it}, \tag{2}$$

with persistent shock $z_{it} = \rho z_{i,t-1} + \eta_{it}$, $z_{i0} = 0$, and iid innovations. Wages of separated workers follow the same process, reduced by the non-parametrically estimated scar profile from Section 2 (twenty post-separation dummies plus constants for second and third separations). Separation occurs at hazard

$$\xi(d,t) = \lambda_0 \Big(e^{\phi t} + \sum_{d=1}^{3} \lambda_d D_d \Big),$$

where λ_0 is a baseline hazard, ϕ captures experience-dependence, and λ_d are loadings on past separations.

We estimate the wage-process parameters by simulated method of moments following Guvenen (2009). Targets include the experience-quadratic coefficients β_1 and β_2, the standard deviations of individual fixed effects and of residual wages

[5] This complements work on the sources of life-time income inequality and risk (Low et al., 2010; Blundell et al., 2008; Huggett et al., 2011; Guvenen, 2009); the distinction is that we directly attribute persistent wage losses to separation rather than absorbing them into general shock processes.

at 5 and 30 years of experience, and three serial-correlation moments $Scorr(1)$, $Scorr(1) - Scorr(2)$, and $[Scorr(2) - Scorr(3)]/[Scorr(1) - Scorr(2)]$.

Parameter estimates appear in Table 1 and the fit is shown in Appendix Table 4. Estimates are comparable to the existing literature.[6] The two columns differ in whether scars are explicitly built in (column 2) or absorbed into the general process (column 1). Once scars are accounted for separately, the AR(1) persistence falls slightly, the variance of individual fixed effects drops, and transitory variance falls. Scars are therefore picking up dispersion that would otherwise be attributed to permanent type or unrelated shocks.

Counterfactual. To quantify how much job loss shapes the distribution, we resimulate using column-2 parameters but with the separation hazard set to zero (Table 2). Removing separations raises 20-year wage growth by 14.7% on average, raises the AR(1) persistence of residual wages from 0.831 to 0.869 (+4.6%), and reduces the standard deviation of estimated individual fixed effects from 0.515 to 0.437 (−17.8%). The contribution to dispersion is concentrated in the permanent component: the standard deviation of residual wages at 5 and 30 years of experience moves by less than 7 percentage points, while the fixed-effect variance moves by more than twice as much. This pattern is consistent with the persistent shape of the scar profile in Figure 2: losses that endure for two decades show up econometrically as worker-level fixed heterogeneity rather than as recurring shocks.

For context, the size of the displacement contribution is large relative to typical estimates of life-cycle income risk. Huggett et al. (2011) attribute roughly 60% of life-cycle inequality to initial conditions and the remainder to shocks; our estimates suggest that displacement alone, treated as a separate channel rather than absorbed

[6] For example, Floden and Linde (2001) use GMM on PSID data and find $\rho = 0.9136$ versus our $\rho = 0.9213$, $\sigma_\eta = 0.206$ versus our 0.2709, and $\sigma_\alpha = 0.2052$ versus our 0.2120; differences reflect our inclusion of iid transitory shocks and a longer sample.

into the general AR(1) process, accounts for nearly a fifth of the dispersion in worker fixed effects. Job loss is therefore quantitatively first-order for both life-cycle wage growth and cross-sectional inequality, not a tail event affecting a small subset of workers, and a wage process that does not include it explicitly will tend to overstate the role of initial conditions and persistent shocks.

4 What Minimum Ingredient Yields Persistent Scars?

We now use a parsimonious learning-by-doing framework to examine which features a reduced-form wage process requires to jointly match the empirical scar profile and the life-cycle moments of the never-separated reference group. The exercise is illustrative rather than diagnostic: we are not claiming that learning-by-doing is the right structural model of wage growth, and we are not assessing whether richer structural models contain a particular mechanism. Rather, we use the framework as a transparent laboratory in which to ask what minimal modification to a restart-on-the-same-path setup is needed to deliver both the life-cycle wage profile and persistent scarring. The skill structure should be read as a stand-in for any process that produces a concave life-cycle wage profile: the same logic applies to job ladders, employer-component wage processes, and learning models.[7]

4.1 Setup

We follow Ljungqvist and Sargent (1998) in modeling life-cycle wages through a learning-by-doing structure. Workers progress through skill levels $h \in \{h_0, h_1 \dots h_N\}$, starting at h_0. When employed, skills increase by step s with probability α. Wages

[7] The recent displacement literature highlights firm and match components of post-displacement wage losses (e.g., Carrington and Fallick, 2017). Our reduced-form framework is consistent with these mechanisms: a step "up" in the simple framework can equally be interpreted as moving to a higher-paying employer or building match-specific capital. The lessons we draw concern the reduced-form path of wages, not the underlying mechanism.

equal h. Workers separate to unemployment with age-dependent probability $\delta(t)$, and on separation lose share τ of skills with probability γ:

$$h' = \begin{cases} h + s & \text{w.p. } (1 - \delta(t)); \\ \tau h & \text{w.p. } \delta(t)\gamma; \\ h & \text{otherwise.} \end{cases}$$

We consider one-year periods over a 35-year career. We calibrate four parameters $(s, \alpha, \gamma, \tau)$ to PSID-estimated separation hazards together with empirical scar coefficients and life-cycle wage growth moments.

4.2 Four Variants

We consider four variants. The **Baseline** is the setup above with random separation hitting a homogeneous worker. **Selection** restricts the separation hazard to workers below a wage threshold, capturing the idea that separations are concentrated among low-productivity matches (Ljungqvist and Sargent, 1998). **Serial Separations** estimates a hazard that loads on past separations: in the PSID, the first separation raises the second-separation hazard by 2.17×, and so on (see Table 1). This captures the spirit of slippery-ladder dynamics in which the rung accessible after unemployment is more separation-prone than higher rungs. **Lowered Growth** retains random separation but reduces the post-separation skill-gain probability by factor ϕ.

We calibrate each variant by jointly targeting the wage scar (initial and 15-year scar plus PDV) and life-cycle moments (mean wage growth over 5 and 30 years), so that the never-separated reference group matches the data.

4.3 Results

Figure 2 (right panel) shows the fit when targeting both scars and life-cycle moments. Baseline, Selection, and Serial Separations miss the persistent scar: the calibration trades short-run depth for long-run flatness, overstating the impact loss and understating the 10-year loss. Lowered Growth alone delivers both, a persistent scar and a 30-year wage profile close to the data (Table 3).

The variant-specific failure modes are instructive. **Baseline** fails for the reason illustrated in Figure 1: with random separation and access to the same growth process, displaced workers re-enter on the steepest part of the concave profile and recover in roughly a decade, regardless of how much skill they lose at separation. Calibrating τ down to deepen the impact scar simply makes the recovery sharper. **Selection** adds depth to the scar (separated workers are drawn from below the wage threshold and have further to fall) but each of them then re-enters the same growth process and the same concavity argument applies; at the calibrated parameters, the 10-year scar is essentially zero. **Serial Separations** achieves persistence through the cumulative effect of repeat displacement, but the empirically estimated hazard loadings ($\lambda_1 = 2.17$, $\lambda_2 = 1.15$, $\lambda_3 = 1.49$) are too small to deliver a -0.09 scar at year 10; matching it would require dramatically more slippery lower rungs than the PSID separation history shows. **Lowered Growth** succeeds because it does the only thing concavity allows: it changes the post-separation trajectory itself, so the gap to the never-separated reference group does not close along the early-career steep portion.

The general lesson is that a reduced-form wage process which leaves separated workers on the same growth path mechanically generates fast recovery, regardless of how the depth of the impact scar is calibrated. Within this parsimonious framework, the minimum ingredient for matching both the scar and the life-cycle reference profile is some channel that lowers post-separation growth on net.

4.4 Discussion

Relation to Ljungqvist and Sargent (1998). Our notion of the scar differs from the original Ljungqvist and Sargent (1998) setup. Following the empirical literature, we define the scar relative to a never-separated reference group rather than relative to the worker's own pre-separation wage. The reference-group definition matters: even with no human-capital loss at separation, a worker who restarts on the same wage path will appear scarred initially because the reference group keeps moving up the steep portion of the concave profile. The mechanical scar from this alone is short-lived (within roughly the time it takes to traverse the steep portion), so persistence requires something more, as the four variants make clear.

Selection. Selection of low-wage workers into separation does help match the depth of the scar: a worker drawn from below the wage threshold has further to fall. But once she returns to the same growth process, the same concavity argument applies: the steep portion of the profile lets her catch up. To deliver persistence, selection has to be on a dimension that *also* affects post-separation growth, which is what the selection-on-types stories in Burdett et al. (2020) and Michaud (2018) achieve.

5 Implementation: A Portable Wage Process

For applied researchers who want a wage process incorporating realistic job-loss risk without taking a stand on the underlying mechanism, we offer two implementation options. Both are designed for use inside a value function with a minimal state space; they differ in how exactly they reproduce the empirical scar profile and in how many states they require.

Option 1: Empirical AR(1) with three state variables. The first option follows the standard Guvenen (2009) form in equation (2) and uses three state variables: experience, the persistent shock z_{it}, and the number of past displacements (capped at three). The transitory shock ϵ_{it} is drawn iid each period and need not be carried as a state. Calibrated parameter values from column 2 of Table 1 are: experience returns $\beta_1 = 0.0237$ and $\beta_2 = -0.0007$, AR(1) persistence $\rho = 0.8996$, innovation standard deviation $\sigma_\eta = 0.3146$, transitory standard deviation $\sigma_\epsilon = 0.1608$, and individual fixed-effect standard deviation $\sigma_\alpha = 0.1334$. The separation hazard is $\xi(d,t) = \lambda_0\big(e^{\phi t} + \sum_{d=1}^{3} \lambda_d D_d\big)$ with baseline $\lambda_0 = 0.96$ and loadings $\lambda_1 = 2.17$, $\lambda_2 = 1.15$, $\lambda_3 = 1.49$ (Table 1). Wages of separated workers are reduced by the nonparametric empirical scar profile from equation (1): a worker with $k \in \{1, 2, 3+\}$ past separations carries the estimated $D1_{i,t-n}$ profile from her most recent separation plus the estimated constants from each higher-order separation, reported in the Appendix. This option matches the empirical scar profile exactly.

Option 2: Wage ladder with two state variables. The second option uses the calibrated Lowered Growth specification from Table 3 and requires only two state variables: the current rung on the wage ladder and an indicator for whether the worker has ever been separated. Workers progress through skill levels with step size $s = 0.199$ and gain probability $\alpha = 0.43$. On separation, they retain $\tau = 0.79$ of their skills and from that point on face a reduced gain probability of $\alpha(1-\phi) = 0.43 \times 0.65 \approx 0.28$. The separation hazard can be the same as in Option 1 or a simpler exogenous hazard, depending on the application. This option is less exact than Option 1 but, as Table 3 shows, delivers a 10-year scar of -0.07 against -0.09 in the data and a 30-year wage profile close to the data, all with one fewer continuous state variable than Option 1. For applications where state-space size is binding, this is a workable approximation.

6 Conclusion

Job loss carries persistent wage consequences. We document that these scars are quantitatively first-order for the wage distribution: separations account for 14.7% of average life-cycle wage growth and 17.8% of cross-sectional wage dispersion. We use a parsimonious learning-by-doing framework to draw out a simple lesson about reduced-form wage processes: because the life-cycle wage profile is concave, restarting displaced workers along the same path mechanically delivers rapid catch-up. Reproducing the persistent scar in the data requires that post-separation growth be lower on net, regardless of which underlying mechanism produces that pattern. We provide a calibrated wage process that builds in this feature directly, intended for applied researchers who need a quantitatively realistic income process incorporating job-loss risk.

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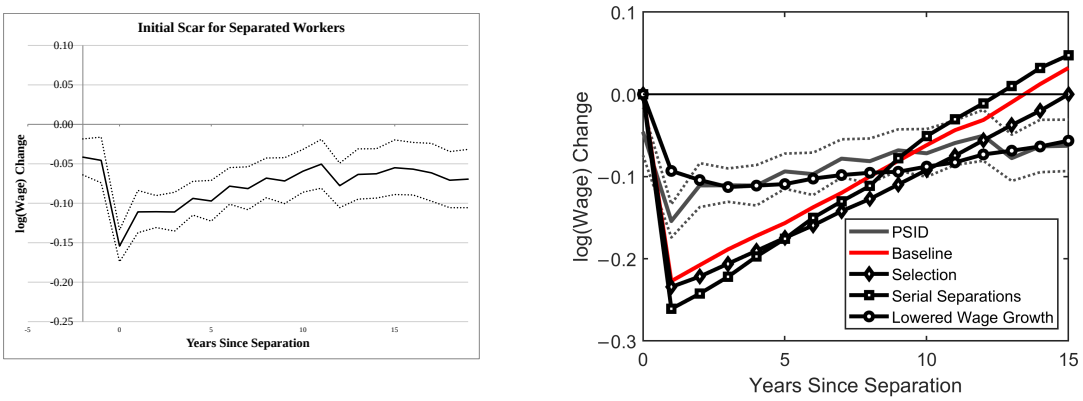
Figure 2: Estimated Wage Scar and Model Fit



Left panel: The solid line indicates the values for $\exp(\beta_{t-n})$ from equation 1 with log hourly earnings as the dependent variable. The dashed lines indicate the 95% confidence intervals. *Right panel:* "Baseline" is the learning-by-doing model of human capital. "Selection" is a modification of Baseline with only low-wage workers facing a separation hazard. "Serial Separations" is a modification of Baseline where the probability of future separations increases after each separation. "Lowered Wage Growth" is a modification of Baseline where workers face reduced wage growth after separation. PSID estimates show the 95% confidence interval.

Table 1: Empirical Wage Model Parameters- Estimates from PSID

Parameter	**No Separations**	(Std. Err.)	**With Separations**	(Std. Err.)
Return to Exp	0.0218	(0.0010)	0.0237	(0.0013)
Return to Exp^2	-0.0006	(0.0000)	-0.0007	(0.0000)
AR(1) persistence (ρ)	0.9213	(0.0112)	0.8996	(0.0333)
std AR(1) innov. (σ_η)	0.2709	(0.0063)	0.3146	(0.0047)
std transitory shock (σ_ϵ)	0.2505	(0.0034)	0.1608	(0.0077)
std permanent level (σ_α)	0.2120	(0.0146)	0.1334	(0.0049)
Initial Separation Hazard (λ_0)	0.0	(n.a.)	0.9582	
Additional Separation Hazards				
After One Separation (λ_1)	0.0	(n.a.)	2.1686	(0.1308)
After Two Separations (λ_2)	0.0	(n.a.)	1.1495	(0.1002)
After 3+ Separations (λ_3)	0.0	(n.a.)	1.4888	(0.1551)

Note: Parameter estimates from simulated method of moments applied to the wage process in equation (2). Column "No Separations" estimates the wage process without a separate separation channel; Column "With Separations" adds the empirically estimated scar profile and separation hazard. The separation hazard is $\xi(d,t) = \lambda_0\big(e^{\phi t} + \sum_{d=1}^{3} \lambda_d D_d\big)$, where d indexes prior separations.

Appendix

Table 2: Role of Separation in the Wage Process

Counterfactual Simulation-Turn off Separation			
Statistic	**No Separation**	**Separation**	**Effect of Sep**
	(Std. Err.)	(Std. Err.)	
20 yr wage growth (mean)	0.395	0.340	-14.7%
	(0.024)	(0.021)	
Resid. Wages, 5 yr Exp (std)	0.483	0.510	+5.6%
	(0.012)	(0.011)	
Resid. Wages, 30 yr Exp (std)	0.496	0.529	+6.7%
	(0.014)	(0.012)	
Wages (Scorr(1))	0.831	0.869	+4.6%
	(0.004)	(0.004)	
Individ. Fixed Effects (std)	0.437	0.515	+17.8%
	(0.011)	(0.012)	

The counterfactual uses the model parameters from the estimation of the model with separation, but then sets separation hazard to zero.

Table 3: Parameter Estimates and Model Fit: Four Variants

Panel A: Parameter Estimates

Parameter Estimates	Baseline	Selected Separation	Correlated Separation	Lowered Growth
Skill Step Size (s)	0.063	0.045	0.116	0.199
Skill Gain Prob. (α)	0.49	0.42	0.41	0.43
Skill Loss Prob. (γ)	0.51	0.02	0.77	0.67
Percent Skills Lost (τ)	0.01	0.12	0.54	0.79
% Reduction in skill growth (ϕ)				0.35

Panel B: Model Fit

Moment	Data	Baseline	Selected Separation	Correlated Separation	Lowered Growth
5 year wage growth (mean)	0.35	0.17	0.11	0.24	0.38
30 year wage growth (mean)	1.10	0.91	0.78	0.99	1.11
30 year wage dispersion (stdev)	0.40	0.09	0.08	0.24	0.47
Initial wage scar (%)	-0.15	-0.23	-0.23	-0.26	-0.11
10 year wage scar (mean)	-0.09	-0.01	-0.03	0.01	-0.07

Note: Panel A reports calibrated parameters of the four variants jointly targeting scar coefficients (initial scar, 15-year scar, PDV) and life-cycle moments (5-year and 30-year mean wage growth). s is the skill step size; α is the per-period probability of moving up a step when employed; γ is the probability of skill loss conditional on separation; τ is the share of skills retained on separation when the loss shock hits; ϕ is the proportional reduction in α for separated workers (Lowered Growth only). Panel B reports implied moments under this calibration; 30-year wage dispersion is not targeted but reported. Only the Lowered Growth specification jointly delivers the persistent scar (10-year scar near -0.09) and the life-cycle profile.

Table 4: Empirical Wage Model: Fit to Additional PSID Statistics

Empirical Wage Model: Estimation Fit			
Moment	**Data**	**No Separations**	**With Separations**
Resid. Wages, 5 yr Exp (std)	0.510	0.510	0.510
Resid. Wages, 30 yr Exp (std)	0.531	0.521	0.529
Wages (Scorr(1))	0.933	0.877	0.869
Wages (Scorr(1)-Scorr(2))	0.039	0.139	0.167
Wages ($\frac{\text{Scorr(2)-Scorr(3)}}{\text{Scorr(1)-Scorr(2)}}$)	0.866	0.872	0.825
Individ. Fixed Effects (std)	0.483	0.462	0.451

Note: Targeted moments from PSID compared to simulated counterparts under the two parameterizations of Table 1. "Resid. Wages" are residuals from a regression of log wages on a quadratic in experience and a constant. $Scorr(n)$ is the n-th order serial correlation of residual wages. Both specifications fit the targets comparably; the difference between them is whether the persistent wage consequences of separation are absorbed into the general wage process (Column 1) or modeled explicitly via the scar profile and separation hazard (Column 2).